7.3 What Makes a Determinant

WORKSHEET

The question

STEP II 2019, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What does each special matrix in the question do? Work out $\mathbf{JM}$, $\mathbf{MJ}$, $\mathbf{KM}$ and $\mathbf{PM}$ for a general $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

STAGE 1 The identity

Covers: Part (i)

Ask yourself Write $\mathbf{M} = \mathbf{MI}$ and apply f to both sides. You may divide by $\mathrm{f}(\mathbf{M})$ for a suitable $\mathbf{M}$.

STAGE 2 Swapping

Covers: Part (ii)

Ask yourself Work out $\mathbf{J}^2$. What does that tell you about $f(\mathbf{J})^2$? Which of the two possible values is ruled out?

Pause and try Write each of the two given matrices as a product of $\mathbf{J}$, $\mathbf{M}$ and $\mathbf{J}$ in some order. Then apply f.

STAGE 3 Dependent rows

Covers: Part (iii)

Pause and try First take a matrix $\mathbf{B}$ whose two rows are equal. What is $\mathbf{JB}$? Apply f and use Part (ii). Then get the matrix with second row k times the first by multiplying $\mathbf{B}$ by $\mathbf{K}$.

STAGE 4 The shear

Covers: Part (iv)

Ask yourself Compute $\mathbf{P}^2$ and $\mathbf{K}^{-1}\mathbf{P}\mathbf{K}$. For which k are they equal?

Pause and try Apply f to both sides of that equation. You get an equation for $f(\mathbf{P})$ with two solutions. Why can $f(\mathbf{P})$ not be 0? Use $\mathbf{P}\mathbf{P}^{-1} = \mathbf{I}$.

What's really going on

Every multiplicative function is a function of the determinant

The question stops at $f(\mathbf{P}) = 1$, but the argument can be completed. Every invertible matrix is a product of elementary matrices (row swaps, row scalings and shears), and the question shows how f behaves on each. It follows that $f(\mathbf{M}) = \phi(\det \mathbf{M})$ for some function ϕ of a single real number with $\phi(xy) = \phi(x)\phi(y)$. The determinant itself is the case $\phi(x) = x$, but $\phi(x) = x^3$, or $\phi(x) = |x|^{1/2}\operatorname{sign}(x)$, work too. In other words, *the determinant is the only fundamentally new multiplicative function on matrices*. In the language of group theory, the determinant captures everything about the group of invertible matrices that can be seen by a commutative group.

The determinant as area

Geometrically, a 2×2 matrix sends the unit square to a parallelogram, and $\det \mathbf{M}$ is its *signed area*: positive if the orientation is preserved, negative if it is reversed (Figure 7.3.1).

The multiplicative property then has an obvious meaning: doing two transformations in turn multiplies their area scale factors. A row swap reverses orientation, which is why $\det \mathbf{J} = -1$; a shear slides the square into a parallelogram of the same area, which is why $\det \mathbf{P} = 1$; and a matrix with dependent rows squashes the square flat, which is why its determinant is zero. Every step of the question has a picture.

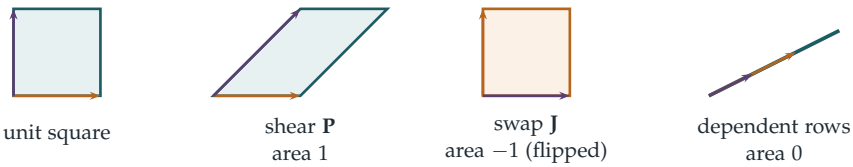


Figure 7.3.1 The determinant as signed area. The images of the two basis vectors (amber, plum) span a parallelogram. A shear preserves area; a swap of the basis vectors reverses orientation; dependent rows (or columns) collapse the square onto a line. Doing two transformations in turn multiplies the signed areas, which is the property $\mathrm{f}(\mathbf{MN}) = \mathrm{f}(\mathbf{M})\mathrm{f}(\mathbf{N})$.

Determinants in higher dimensions

For $n \times n$ matrices the determinant is the signed volume scale factor, and it is characterised by three properties: it is linear in each row, it changes sign when two rows are swapped, and it equals 1 on the identity. From these alone you can derive the familiar expansion formulas, and Gaussian elimination gives an efficient way to compute it. The determinant appears in the change-of-variables formula for multiple integrals (the *Jacobian*), in the formula for the inverse of a matrix, and, as we saw in Section 5.4, in the discriminant of a polynomial.

Try this next

1 **Other examples.** Show that $\mathrm{f}(\mathbf{M}) = (\det \mathbf{M})^3$ satisfies all the conditions of the question, and that $\mathrm{f}(\mathbf{M}) = |\det \mathbf{M}|$ satisfies the multiplicative property but not $\mathrm{f}(\mathbf{J}) \neq 1$.

2 **Column operations.** Show that f is also zero on any matrix whose second column is a multiple of the first.

3 **Scalings.** Let $\mathbf{K}_k = \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$. Show that $\mathrm{f}\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix} = \mathrm{f}(\mathbf{K}_k)$, and that $\mathrm{f}(\mathbf{K}_k)\mathrm{f}(\mathbf{K}_l) = \mathrm{f}(\mathbf{K}_{kl})$.

4 **Reduction to the determinant.** HARDER Show that every invertible 2×2 matrix can be written as a product of matrices of the forms $\mathbf{J}$, $\mathbf{K}_k$ and $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$. Deduce that $\mathrm{f}(\mathbf{M})$ depends only on $\det \mathbf{M}$.

Further reading

- **Sheldon Axler, *Linear Algebra Done Right*** (Springer; free online). Defines the determinant from its properties, in the spirit of this question.
- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Chapter 5 develops determinants from three rules.
- **3Blue1Brown, "The determinant"** (video, free online). The area interpretation, beautifully animated.

The full guided solution is in Section 7.3:
https://beyondstep.keithmcnulty.org/ch07/determinant-axioms.html